

Asymmetric dependence between the prospect theory value and stock returns in India: a quantile regression approach

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Abstract

Purpose – Motivated by the evidence highlighting the role of sentiments and cognitive biases in investors' decision-making, this study examines a novel behavioral finance-based asset pricing model incorporating the prospect theory framework in the Indian equity market. Specifically, the study's primary objective is to investigate the importance of Prospect Theory Value (PTV) in the cross-sectional pricing of stocks.

Design/methodology/approach – The empirical findings rely on data taken from NIFTY 500 and BSE S&P 500 stocks, encompassing daily, weekly and monthly observations. The analysis employs diverse statistical techniques, including Ordinary Least Squares (OLS), Fama–Macbeth Cross-section Regressions, Panel Fixed Effect and Quantile Regression.

Findings – The study demonstrates an asymmetric association between PTV and subsequent stock returns. The findings maintain their robustness even when factoring in stock-specific attributes such as market capitalization and book-to-market ratio, market beta and indicators related to lottery-like behavior such as skewness and MAX. This observed pattern persists when analyzing data at various frequencies, including daily, weekly and monthly intervals. Loss aversion behavior dominates among Indian equity investors, contrary to lottery preferences in the US equity market.

Originality/value – As far as the authors are aware, the study is the first to introduce a new behavioral finance-motivated stock return predictor (PTV) in the Indian stock market. The study also marks the pioneering use of a novel method that evaluates the predictability of PTV across various sections of the conditional return distribution using quantile regression.

Keywords Asset pricing model, Behavioral biases, Prospect theory, Quantile regression, Risk preferences
Paper type Research paper

1. Introduction

The investment decision-making field has long relied on Expected Utility Theory (EUT, henceforth) to understand how decisions are made in risky and uncertain situations (Tversky, 1975). Classical finance theories, such as Modern Portfolio Theory and Capital Asset Pricing Model (CAPM, henceforth), have relied on the rationality assumptions of EUT to provide insights into investor decision-making, portfolio management, and asset pricing (Markowitz,

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1952; Sharpe, 1964). However, the empirical literature has increasingly emphasized significant deviations in investor behavior from the EUT assumptions, leading to challenges in explaining various stylized facts related to stock returns (Barberis and Huang, 2008).

Consequently, there is a growing interest from academia for alternative approaches that better capture the intricacies of investor decision-making. These alternative approaches do not make the assumptions of rationality intrinsic in EUT and draw from the stream of psychology. Thus, they offer more accurate explanations for observed market phenomena. Of these, one prominent alternative is Prospect Theory (PT, henceforth) by Tversky and Kahneman (1975), which integrates reference-dependent risk preferences, loss aversion, and decision weights within a framework for making choices in uncertain and risky choices, providing a more realistic perspective on judgment in settings of risk and uncertainty.

Despite PT's recognition as the most reliable alternative to the models of rational equilibrium, most studies are limited to its theoretical and experimental front. Barberis *et al.* (2016) conducted a notable empirical test of PT within the US equity market. They suggest that, at first, investors mentally observe security by its past returns and then assess this pattern as the PT suggests. Their study identifies a robust connection between the stock's Prospect Theory Value (PTV, henceforth) for the preceding month and its subsequent stock price performance. Similar patterns of PTV's link to return predictability have since been observed in various markets, including mutual funds (Akbas and Genc, 2020), foreign share market (Wang *et al.*, 2021), cryptocurrencies (Chen *et al.*, 2022; Thoma, 2021; Yadav, 2024) and corporate bond market (Zhong and Wang, 2018).

Our study examines two hypotheses. First, the PTV is a significant predictor of stock returns, and second, there are variations in investors' risk preferences towards stocks that perform below or above the mean, with potentially heightened irrationality during extreme market conditions. To test these hypotheses, we employ a comprehensive methodology combining standard conditional mean models (including Pooled OLS, Fama-MacBeth cross-sectional regressions, Panel fixed effects) and QR in the second stage of Fama and Macbeth's (1973) cross-section regression [1]. Utilizing this approach allows a comprehensive exploration of how the association between returns and PTV evolves across different levels of conditional returns, revealing the dynamic link between PTV and future returns.

We contribute to the literature and examine the predictions of PTV in the context of the Indian market, a significant emerging economic landscape. The current market capitalization of India amounts to \$3.31 trillion, positioning it as the fifth most valuable country in terms of market capitalization [2]. Another motivation behind this research stems from the unique attributes of the investor base in the Indian equity market, which features a significant proportion of retail investors compared to other major markets such as the US [3]. This composition makes the market more susceptible to behavioral and cognitive factors, which are the focus of this study [4]. Moreover, few studies have examined the behavioral bias and risk preferences of Indian investors (Aziz and Ansari, 2018). Consequently, PT emerges as a valuable framework for understanding investors' risk attitudes and preferences and revealing vital information on stock return dynamics. Nevertheless, there is no empirical evidence that illustrates the potential influence of PT in explaining the behavior of Indian stocks. Our research intends to meet this gap.

Furthermore, our study addresses the need for replication in economic studies, as highlighted in the literature [5]. Our study satisfies the conditions of scientific replication using different samples, periods, and statistical models from the existing literature (Hamermesh, 2007). Furthermore, similar to any findings in empirical research, the adverse relationship between PTV and returns might be a consequence of data snooping (Lo and MacKinlay, 1990) and, hence, potentially limited to the specific dataset. Our study mitigates this concern.

To comprehensively assess the influence of investment horizons on investor behavior and risk preferences, we analyze stock data at various time intervals: daily, weekly, and monthly

[6]. This systematic methodology reinforces the reliability and broader relevance of our research findings. Lastly, our study also delves into which component of the PTV (loss aversion, lottery preferences, and concavity) is most significant in the Indian market, showing which particular behavior is more dominant in Indian equity investors.

Our results unveil a dynamic trend in the relationship between PTV and distinct conditional stock return distribution segments. Lower quantiles demonstrate a positive correlation with PTV, whereas upper quantiles reveal a negative correlation. Investigating the PTV and return dependence on the extreme quantiles, we find that the relationship becomes statistically significant as we go distant from the median quantile and is most pronounced at the distribution tails. We also note that the positive relationship in the lower quantiles is more pronounced than the negative relationship in the upper quantiles. While MAX and skewness are linked to PTV, we establish that the PTV's ability to predict outcomes extends beyond the scope of conventional proxies for lottery-like characteristics. Another concern that PTV might be proxying for short-term reversal is addressed by including reversal as a control variable in our models and by taking a one-month gap between the PTV computation and return, and our results remain the same. One might link our findings to arbitrage limitations. However, it's crucial to recognize that our results should not be attributed to arbitrage constraints because our sample includes explicitly the largest 500 and most informationally efficient securities (NIFTY 500). Further, we also conclude that loss aversion behavior is the most prevalent among Indian investors, highlighting the contrasted risk preferences of Indian investors [7].

We conduct multiple robustness assessments to reinforce our findings. We find a consistent return predictability of PTV to alternative performance measures, alternative PTV estimation, and prediction windows. The relationship is also robust to unfiltered data, a more extensive sample period (Since 2000), and Indian PT parameters suggested by [Rieger et al. \(2017\)](#).

Our study makes substantial contributions to various facets of prevailing literature. First, our study is the first to introduce the new stock return predictor (PTV) rooted in behavioral finance theory within the Indian stock market context, challenging the existing asset pricing models. Second, the paper introduces a QR framework to investigate the predictive capacity of PTV across various sections of the conditional return distribution. This analytical step is novel in the literature and extends the prior analysis, offering valuable guidance for portfolio managers [8] and policymakers in managing market volatility. Third, this study augments the existing research exploring the asset pricing inferences of PT by offering fresh empirical findings within a significant emerging economy context. While the extant studies have investigated PT in the equity markets of the USA and other developed economies, corporate bonds, and foreign exchange markets, our study provides significant "out-of-sample" results concerning the cross-sectional predictability of PTV within the NSE, an important emerging market. Finally, ours is the first study to assess the influence of investment horizons on investor behavior; we analyze stock data at different frequencies: daily, weekly, and monthly.

The study is structured as follows: [Section 2](#) offers a summary of the existing literature. [Section 3](#) outlines the data sources and methodology utilized in this investigation. [Section 4](#) describes the empirical findings. [Section 5](#) reflects on the implications, limitations, and avenues for future research, while [Section 6](#) concludes the study.

2. Literature review and contribution

[Barberis et al. \(2016\)](#) construct an asset pricing model where investors have PT preferences. They conclude that in the US equity market, investors find stocks with high PTV more appealing and, therefore, make them part of their portfolio. This leads such stocks to become overvalued, resulting in lower successive returns. They also suggest that this relationship is

more robust in individual investors and, therefore, in small, illiquid, and highly volatile stocks. Furthermore, examining the estimate of PT in 46 other markets, they found that certain markets didn't conform to PT's predictions; however, they do not provide an account of these countries.

Literature indicates that investors across countries display distinct levels of psychological prejudices due to cultural differentiations (Kumar, 2009). Demographic considerations like religious background also play a principal role in developing the value system that eventually gets reflected in the differences in attitudes of investors towards risk and uncertainty. Consequently, exploring the application of PT to asset pricing in different markets across different asset classes has gained significant traction [9]. In this line, Son and Nguyen (2019) note a positive PTV-returns link in South Korea, which is identified as a collectivist society (Eun *et al.*, 2015). They suggest that Korean investors tend to prefer stocks with lower PTV than the average as they are driven by the irrational belief in the greater upside potential of these stocks, even if they have subpar future returns (Birru and Wang, 2016; Kumar, 2009). Conversely, Eom and Park (2020) show that the negative PTV-return association is sensitive to short-term reversal factors and the fat-tail feature in return distributions. Additionally, demographic factors such as religious associations and ethnic backgrounds are also related to different risk preferences. Chen *et al.* (2024b) explore two submarkets in Thailand that have a highly religious investor base and conclude a negative association between PTV and returns, with loss aversion better capturing biases during crises and bear periods and probability weighting during the bull markets.

Moreover, existing literature also indicates that institutional investors, due to their higher level of sophistication, are less prone to behavioral biases. Wang *et al.* (2021) highlight that the PTV becomes significant in China's B-share market only after domestic retail investors gain access to it. They attribute the shift to the probability weighting component of PT. In another study, prompted by the evidence of the anchoring effects, Chen *et al.* (2024a) unveil a negative PTV-return relationship in the Chinese stock market, particularly in stocks distant from their 52-week highs. They further state that loss aversion performs the most crucial role in the relationship between the anchor factor and PTV. do Nascimento *et al.* (2021) explore various emerging markets, noting variations in the relationship between PTV and returns attributed to cultural distinctions, with Russia, China, and Brazil demonstrating negative correlations, Mexico displaying robust positive correlations, and South Africa exhibiting varied results.

Moreover, considering the prevalence of retail investors in mutual funds, several studies empirically investigate whether PT helps explain mutual fund performance. Analyzing quarterly data for China equity funds, Wang and Han (2023) show a robust positive link between fund-level PTV and subsequent fund flows. Similar findings emerged in studies by Guo and Schoenleber (2020), Han *et al.* (2021), and Gu and Yoo (2021) in the US fund market, and Gupta *et al.* (2022) in the global fund market. Yu *et al.* (2022) attribute this positive correlation in the Chinese market to performance persistence and the price impact hypothesis. Notably, the predictive ability of PTV is not subsumed by existent fund persistence indicators like fund momentum.

Zhong and Wang's (2018) study on US corporate bond returns reveals that, analogous to the equity market, investors favor bonds with high PTV, which results in lower future returns. An analogous observation is made by Chen *et al.* (2022) in the cryptocurrency market. Both studies highlight a stronger PT effect in difficult to arbitrage securities. However, while PW drives the PT effect in the cryptocurrency market, the loss aversion factor is dominant in the corporate bond market of the US, which contrasts with the equity market.

The Indian stock market, boasting a market capitalization of \$3.31 trillion and characterized by significant retail investor presence, presents an exciting opportunity to test

the PT predictions. In their study, [Barberis et al. \(2016\)](#) conclude that PT propositions are more robust in stocks traded by individual investors. Despite various studies exploring PT-related behaviors [such as the disposition effect ([Bharande and Rao, 2020](#)) and the inclination for lottery-like stocks ([Aziz and Ansari, 2018](#))], no study has examined the asset pricing implications of PT in the context of India equity market. Our study fills this gap by conducting a significant out-of-sample test of PT estimates for stock returns in a distinct sociocultural environment and offers a novel understanding of its ability to explain the trading behavior of retail investors. Based on this, we propose to investigate whether PTV can predict stock returns across the cross-section of the Indian stock market.

Our research also adds to the literature by providing a more intricate analysis, using data on daily, weekly, and monthly intervals. This decision stems from the evolving landscape of financial markets and the availability of increasingly granular data. Prior research primarily relied on monthly data to analyze PTV and cross-sections of security returns. [Barberis et al. \(2016\)](#) analysis includes the universe of CRSP stocks from 1926 to 2010. They assume that investors rely on monthly data of the preceding 60 months to form their expectations [10]. They argue that since most of their data originates from the pre-internet period when the investment handbooks were the reference sources of the stocks, only the monthly fluctuations were discernible enough to have an influence over investors' representation of the stock. Following this, the subsequent literature in the emerging markets have also performed their analysis on monthly frequency.

However, today, modern financial markets with limit order book microstructure and electronic communications networks have become extremely fast and capable of handling large volumes of trades. Therefore, the speed and capacity at which the financial markets operate has increased exponentially ([Chordia et al., 2005](#)). Also, in these financial markets, with the availability of computing power, information dissemination has increased rapidly; therefore, not only the monthly, weekly, and daily data but also high-frequency trading data is becoming easily accessible. Utilizing these finer time intervals provides several advantages, including increased data points, which in turn yield more efficient estimations. Additionally, the prevalence of High-Frequency Trading (HFT) has led to swift corrections of market anomalies within short time horizons ([O'Hara, 2015](#)). Therefore, testing for mispricing at the highest possible frequency is advisable to capture and analyze these rapid market adjustments effectively. In this context, [Chen et al. \(2022\)](#) utilize the weekly frequency to investigate cryptocurrency returns, covering the period from 2014 to 2020. In a similar line, [Thoma \(2021\)](#) and [Yadav \(2024\)](#) consider daily frequency in the cryptocurrency market. Examining whether the short-term return predictability of MAX (maximum return) is driven by PTV, [Baars and Mohrschladt \(2021\)](#) utilize daily frequency and consider 30 days of the previous month as the PTV construction window. While intraday trading data is readily available, it is too noisy and might not be very efficient in forming the expectations of the return, construction of variables, and study design. To comprehensively explore the influence of investment horizons on investor behavior and risk preferences, we make a finer compromise and undertake analysis at various time intervals: daily, weekly, and monthly.

Thus, our research addresses a crucial void in the literature by offering insights into how various time intervals influence the pricing consequences of PT in equity markets. By examining PTV at different frequencies, we shed light on how market anomalies and investor behavior manifest over shorter and longer time horizons, thereby enhancing our knowledge of the pricing implications of PT in financial markets.

Moreover, the existing studies examining the PT implications on asset pricing primarily rely on conditional mean estimation to unveil the returns and PTV relationship. Nonetheless, it is widely recognized that the stock returns data show nonlinearities (high skewness and excess kurtosis). These attributes limit the insights obtained exclusively from estimates

formulated on the conditional mean. To address this challenge, we adopt the QR given by [Koenker and Bassett \(1978\)](#) in the second leg of FMB cross-section regression. QR allows a more comprehensive identification of how the association between returns and PTV, as a risk factor, changes across different levels of conditional returns. Thus, our hypothesis centers on potential disparities in investors' risk preferences towards securities that perform below or above the mean. Additionally, we also consider the possibility of sharp heterogeneous risk preferences during periods of exceptionally low or high returns, which can lead to varying degrees of predictability based on PTV. This enhances the contributions of our research, as opposed to exclusively modeling "mean" returns using a least squares approach; we estimate models that characterize quantiles of the conditional distribution of returns as functions of PTV.

3. Data and methodology

3.1 Summary and descriptive statistics

The sample stocks of the study are the NIFTY 500 stocks, representing about 90% of the equity market. Daily data for NIFTY 500 is obtained from the Bloomberg database. By selecting these top 500 stocks, we ensure our findings are not influenced by small, less liquid, and highly volatile stocks. Our dataset spans a more recent period, starting in January 2010 and extending through December 2022.

To reduce the influence of outliers, we winsorize all the variables at the 1st and 99th percentiles. Detailed definitions of each variable are outlined in [Table 1](#). We calculate the PTV, momentum, and reversal using monthly stock returns, while daily returns are employed to formulate variables such as market beta, idiosyncratic volatility, liquidity, and various skewness metrics. The calculation of the PTV variable is detailed in [Appendix 2A](#). [Table 2](#) (Panel A) displays the mean values of these study variables over the years. [Table 2](#) (Panel B) presents the mean of monthly cross-sectional summary statistics over time. Further, [Table 2](#) (Panel C) depicts the correlation coefficients between variables. PTV is

Variable	Definition
PTV	PTV of the previous three years' return distribution (Barberis et al., 2016 ; Tversky and Kahneman, 1992)
Beta	Beta is computed using the slope coefficient of regression of the stock's excess return on the market excess return and also the lag and lead of the market portfolio (Dimson, 1979 ; Scholes and Williams, 1977)
Bm	The book-to-market ratio in month $t-1$
Size	Natural log of stocks' market capitalization
Momentum	Momentum is the cumulative return from $t-12$ to $t-2$ months
Reversal	Reversal is the cumulative return from $t-1$ month
Ivol	Idiosyncratic volatility is the volatility of a stock's previous month's daily return
Max	Max is the stock's maximum one-day return in the previous month (Bali et al., 2011)
Min	Min is the stock's minimum one-day return in the previous month (Bali et al., 2011)
Skewness	Skewness is the skewness of the previous month's daily return of a stock
Iskew	Idiosyncratic skewness is the stock's own skewness. It is computed as the skewness of the residual term of regression on market excess return over risk-free rate and squared market excess return (Harvey and Siddique, 2000)
Co-skewness	Co-skewness is also referred to as systematic skewness. The slope coefficient of squared market excess return is taken as the co-skewness of the stock (Harvey and Siddique, 2000)
Note(s): Definitions of various firm-level and macroeconomic variables used in the study are provided in the table	
Source(s): Created by authors	

Table 1.
Variable descriptions

Table 2.
Descriptive statistics
and correlation matrix

Panel A: Variable summary														
Year	Mean cap	Median cap	Mean volume	Median vol	PTV	Beta	Bm	Ivol	Max	Min	Skewness			
2013	163.02	32.89	347.65	346	−0.071	0.898	−0.496	0.02	0.049	−0.042	0.232			
2015	245.09	70.66	369.41	368.35	−0.06	1.074	−1.018	0.02	0.05	−0.045	0.179			
2017	295.86	99.09	385.77	385.89	−0.053	1.134	−1.186	0.017	0.046	−0.033	0.394			
2019	298.88	102.28	374.58	372.56	−0.068	1.12	−0.986	0.018	0.046	−0.04	0.217			
2021	415.92	159.73	407.68	406.92	−0.078	0.969	−1.371	0.02	0.053	−0.041	0.293			
2022	458.88	188.36	403.25	402.47	−0.073	1.032	−1.377	0.019	0.049	−0.043	0.165			
Panel B: Summary statistics														
	Return	PTV	Beta	Size	Bm	Mom	Reversal	Ivol	Illiq	max	Min	Skewness	Iskew	Co-skew
Min	−0.299	−0.208	−2.105	7.774	−3.556	−0.831	−0.3	0.007	0.003	0.012	−0.145	−1.83	−1.69	−0.834
1st Qu	−0.049	−0.088	0.371	10.559	−1.772	−0.094	−0.049	0.013	0.076	0.03	−0.051	−0.237	−0.137	0.47
Median	0.011	−0.061	0.999	11.472	−1.085	0.139	0.011	0.018	0.166	0.043	−0.036	0.221	0.301	0.922
Mean	0.015	−0.066	1.069	11.531	−1.046	0.167	0.015	0.02	0.224	0.05	−0.042	0.24	0.326	0.987
3rd Qu	0.077	−0.039	1.699	12.512	−0.302	0.403	0.077	0.023	0.303	0.061	−0.026	0.7	0.769	1.433
Max	0.343	0.014	4.741	15.111	1.32	1.387	0.343	0.052	1.086	0.172	−0.011	2.433	2.411	3.366
SD	0.109	0.041	1.169	1.487	1.034	0.408	0.109	0.009	0.208	0.029	0.024	0.78	0.747	0.781
SK	0.156	−0.845	0.336	0.041	0.035	0.385	0.154	1.335	1.76	1.714	−1.823	0.12	0.132	0.471
Kurt	3.905	4.172	4.021	2.915	2.572	3.524	3.904	5.066	6.655	6.648	7.457	3.462	3.439	3.509
Panel C: Correlation matrix														
	Return	PTV	Beta	Size	Bm	Mom	Reversal	Ivol	Illiq	max	Min	Skewness	Iskew	Co-skew
Return	1													
PTV	0.097***	1												
Beta	−0.020***	−0.113***	1											
Size	−0.027***	0.186***	−0.033***	1										
Bm	−0.025***	−0.455***	0.117***	−0.326***	1									
Mom	0.039***	0.367***	−0.012*	0.036***	−0.272***	1								
Reversal	0.015**	0.104***	−0.015**	0.007	−0.077***	0.317***	1							
Ivol	0.209***	−0.220***	0.165***	−0.331***	0.167***	0.048***	0.024***	1						
Illiq	0.196***	−0.159***	0.080***	−0.251***	0.153***	−0.043***	−0.053***	0.513***	1					
max	0.405***	−0.150***	0.201***	−0.254***	0.135***	0.016***	−0.003	0.828***	0.421***	1				
Min	0.326***	0.233***	−0.240***	0.156***	−0.174***	−0.003	0.007	−0.629***	−0.312***	−0.366***	1			
Skewness	0.330***	0.033***	−0.012*	−0.120***	−0.019***	0.006	−0.002	0.170***	0.046***	0.526***	0.429***	1		
Iskew	0.287***	−0.013	0.028***	−0.104***	0.013**	0.023***	0.020***	0.167***	0.027***	0.468***	0.255***	0.753***	1	
Co-skew	−0.031**	−0.155***	0.622***	0.015**	0.173***	0.029***	0.004	0.193***	0.089***	0.268***	−0.330***	−0.031***	0.011*	1

Note(s): Panel A reports the average and median market capitalization and trading volume for the years 2013–2022 (alternate year), along with the mean values of other relevant variables considered in the study. Panels B and C present the time-series averages of a set of monthly cross-sectional summary statistics. Panel B reports the summary statistics, including mean, median, minimum, maximum, first and third quartiles, skewness, and kurtosis. Meanwhile, Panel C reports the correlation matrix computed from the cross-section month by month and the time-series averages of the monthly statistics

Source(s): Created by authors

positively related to past performance measures like momentum and reversal, whereas it is negatively associated with dispersion measures like volatility and idiosyncratic volatility. It is also positively related to past skewness. In addition, we standardized all variables by deducting their individual means and dividing them by their corresponding standard deviations (SD or SDs, henceforth) to compare each independent variable's relative influences on the dependent one. Further, the return on the NIFTY 500 index serves as a representative of market return [11].

3.2 Methodology

In panel data sets like ours, researchers have devised various ways to address the potential biases in the standard errors. We begin our analysis using the standard conditional mean models. However, recognizing the limitations of each of these approaches, we utilize multiple methods (Pooled OLS, FMB cross-section regressions, Panel fixed effects, and QR) as we move toward a more robust estimate. Pursuant to Barberis *et al.* (2016), in all the conditional mean models, our hypothesis remains the same, that is, PTV is a robust predictor of stock returns. While Pooled OLS offers simplicity, it falls short in capturing important features of stock return distributions due to its assumptions. OLS assumes errors to be independent and identically distributed, which is frequently violated in the panel data, and thus, the estimates turn out to be biased.

To address this, the FMB approach is widely utilized in asset pricing literature for estimating regression estimates and standard errors, especially when residuals are not independent. We further test our hypothesis using FMB regressions, which also enable us to compare our findings with the extant literature. In this approach, the T cross-sectional regressions are performed, and the mean of these T estimates provides the coefficient estimate. In cases where only a time effect exists, FMB provides unbiased and more efficient estimates compared to pooled OLS. However, in the company of a firm effect, FMB variance estimates can be too small (Cochrane, 2001), leading to underestimated standard errors similar to pooled OLS. By conducting cross-sectional regressions for each time interval, FMB accounts for the time effect through the intercept, significantly reducing residual variability compared to pooled OLS and, consequently, enhancing efficiency.

However, in the existence of a fixed firm effect, both pooled OLS and FMB may not yield the best unbiased estimates (Petersen, 2009). To address this concern of heterogeneity across firms, the literature commonly turns to fixed effects panel data estimation [12]. This approach effectively addresses concerns related to heteroscedasticity and endogeneity concerns due to omitted variables.

In this context, we test the following hypothesis, **H1**: PTV is a significant predictor of stock returns, and estimate the below equations:

$$r_{i,t+1} = \alpha + \alpha_1 PTV_{i,t} + \delta X_{i,t} + u_{i,t+1} \quad (1)$$

$$r_{i,t+1} = \gamma_{0,t} + \gamma_{1,t} PTV_{i,t} + \delta X_{i,t} + u_{i,t+1} \quad (2)$$

$$r_{i,t+1} = \beta_1 PTV_{i,t} + \delta X_{i,t} + u_{i,t+1} \quad (3)$$

We estimate Equation (1) using pooled OLS. In Equation (2), we examine the link between PTV and expected stock returns using FMB cross-section regressions. Every month, we perform a cross-sectional analysis where we regress the returns of individual stocks for that month against the lagged PTV from the preceding month ($PTV_{i,t}$), alongside other explanatory factors ($X_{i,t}$). Subsequently, we provide the time-averaged coefficients for these independent variables. Finally, Equation (3) represents the Panel fixed effects model, and robust standard errors (HAC) are estimated using Arellano's method (1987). $X_{i,t}$ includes stock-specific control variables (as listed in Table 1).

A significant loss of efficiency in the conditional mean models occurs because the least squares models concentrate on the mean as a measure of location. This approach overlooks evidence about the tails of the distribution. Moreover, extreme outliers can substantially skew the estimated results. Thus, acknowledging the limitations of existing literature relying heavily on conditional mean models, we adopt a more robust methodology, i.e., QR.

QR has found widespread application within the field of finance, as it accounts for stylized facts, such as high skewness, kurtosis, and fat tails, thereby providing a better understanding of the asymmetric features of the stock return distribution. It is particularly popular in understanding the Value at Risk (VaR) measure (Merlo *et al.*, 2021). Ma and Pohlman (2008) examine the comprehensive impact of factors on security returns, uncovering significant variations in factor effects across various return quantiles. Further, exploring the dynamic idiosyncratic risk-return relationship, Nath and Brooks (2015) found increasing slope parameters of idiosyncratic risk and highly robust estimates in the tails of the return distribution. Barnes *et al.* (2002) apply this method to study the role of one-factor CAPM beta risk in describing returns and demonstrate the robustness of QR over the least square method. Subsequently, Aziz and Ansari (2018) utilize QR to model the Min effect in their study concerning the cross-sectional analysis of stock market returns. Chuang *et al.* (2009) also revealed a heterogeneous and dynamic stock market index returns and trading volume relationship across different quantiles. These studies collectively indicate that the structure of return-risk factors is diverse. Consequently, models with constant parameters across states, such as conditional mean-based least square models (e.g., pooled OLS), may provide misleading conclusions.

Thus, our study is the first to model the asymmetric link between PTV and stock returns as we test the following hypothesis, **H2**: Heterogeneity exists in the PTV-return relationship across distinct quantiles of the return distribution. QR model that we estimate is:

$$Q_{r,t+1}(\tau|PTV_{i,t}) = \alpha(\tau) + \beta_1(\tau)PTV_{i,t} \quad (4)$$

The conditional quantile function for the “ τ -th” percentile ($0 < \tau < 1$) of the return $r_{i,t+1}[Q_{r,t+1}(\tau|\dots)]$ is represented as a linear equation involving the past PTV, encompassing a range of τ values (e.g., $\tau = 0.05, 0.10, 0.15, \dots, 0.90, 0.95$). The coefficients $\beta_1(\tau)$ may be interpreted as PTV parameters specific to each quantile, which constitute the core of this study.

We expand the Equation (4) to include additional explanatory variables in Equation (5). The modified model is as follows:

$$Q_{r,t+1}(\tau|PTV_{i,t}, \dots, Coskew_{i,t}) = \alpha_0(\tau) + \beta_1(\tau)PTV_{i,t} + \dots + \beta_k(\tau)Coskew_{i,t} \quad (5)$$

These models are implemented using the “*quantreg*” package in R language. We compute the bootstrapped standard errors and variance-covariance matrices that are robust to serial correlation, heteroskedasticity, and any general dependence between the regressors and the error term.

4. Empirical results

4.1 Least square regressions results

We begin by examining the relationship between PTV and returns using various conditional mean regression techniques, addressing our first hypothesis (H1). The results are detailed in Table 3. While the pooled OLS analysis, consistent with Barberis *et al.* (2016), suggests an inverse link between PTV and subsequent excess returns, it lacks statistical significance. However, since the model makes certain assumptions about the error term distribution, the estimates often turn out to be biased and, hence, unreliable.

	<i>N</i>	PTV	Beta	Size	Bm	Mom	Reversal	Illiq	Ivol	Max	Min	Skew	Iskew	Coskew	<i>Adj R</i> ²	<i>F</i> value
OLS	41,689	−0.0082 (−1.206)	−0.0461*** (−6.807)	−0.0536*** (−9.414)	0.021*** (3.71)	0.036*** (5.53)	−0.008 (−1.299)	0.018*** (2.59)	−0.037** (−2.44)	0.043*** (2.73)	−0.034*** (−3.075)	−0.008 (−0.718)	0.026*** (3.39)	−0.020*** (−2.677)	0.98%	27.53***
PLM	41,689	−0.0189 (−0.837)	−0.0012** (−2.141)	−0.0268*** (−14.497)	0.002 (1.07)	0.016*** (8.30)	0.013** (2.21)	2.994 (0.93)	−0.329* (−1.666)	−0.03 (−0.544)	−0.113** (−2.284)	0.003* (1.92)	0.002* (1.96)	0.001 (−0.28)	0.51%	59.11***
FMB	41,689	0.0206* (1.70)	−0.0156* (−1.686)	−0.0694*** (−4.940)	0.0094 (0.636)	0.0438*** (3.485)	0.0073 (0.834)	0.0025 (0.379)	−0.0551*** (−3.674)	0.008 (0.555)	−0.0278** (−2.01)	0.0143 (1.172)	0.0052 (0.721)	0.001 (0.083)	10.21%	4.94***

Note(s): (1) “*N*” indicates the number of observations. (2) *, **, and *** indicate statistical significance at 10%, 5%, and 1% levels

Following [Equations \(1\),\(2\) and \(3\)](#), results for OLS, panel fixed effects, and Fama-Macbeth cross-section regressions are presented. We report the coefficients of PTV and other control variables, *t*-statistics (below), *adjusted* – *R*², and *F*-values

Source(s): Created by authors

Further, transitioning to a more robust estimation procedure, the FMB cross-section regression, known for accounting for time effects and widely utilized in asset pricing literature, revealed a positive association between PTV and future stock returns. Table 3 presents the averaged coefficients over time derived from the regressions of future stock returns with respect to PTV. A positive dependence between future excess stock returns and PTV is statistically significant even after including various firm-level characteristics. The positive association aligns with Son and Nguyen (2019) in the South Korean stock market and do Nascimento *et al.* (2021) in Mexico, which they attribute to the differences in the sociocultural structure of the society and the value system of the individual. However, considering heterogeneity across firms, the results from FMB estimates should be looked at with certain caution (Petersen, 2009). This is because, in the presence of fixed firm effects, standard error estimates are significantly biased downward, potentially compromising the robustness of the observed positive association between PTV and stock returns.

Moving to the fixed effects panel data estimation, which addresses heterogeneity at the firm level, we report a negative link between PTV and stock returns. However, the relationship is not statistically significant. Our model controls for both firm fixed effect and time effect. This approach helps address concerns related to endogeneity and omitted variable bias. Robust standard errors (HAC) are estimated using Arellano's method (1987). Thus, in panel datasets like ours, where researchers have employed various approaches to address the possible biases in the standard errors among conditional mean models, fixed effects panel data estimation offers the most robust analysis.

A statistically significant positive PTV-returns relationship in the Indian market contrasts the findings of Barberis *et al.* (2016). It underscores the importance of cultural influences in shaping individuals' risk preferences. For instance, several studies document that investors from different countries exhibit varied levels of psychological biases due to cultural differences [for example, Chui *et al.* (2010) and Kumar (2009)]. Demographic factors like age and sociocultural factors like religious beliefs vary across countries, resulting in diverse value systems. Consequently, this leads to deviations in how individuals make investment decisions (Dohmen *et al.*, 2011; Tekçe and Yilmaz, 2015). The models employed in the study accurately model returns and assess the PTV at the mean of the conditional return distribution. In other words, they analyze and test assets that conform to the behavior predicted by the mean.

In this context, QR doesn't assume specific error term distributions, handles heteroscedasticity, and is robust to model misspecifications, providing a more precise and comprehensive analysis (Koenker and Bassett, 1978). Thus, while Barberis *et al.* (2016) findings contribute valuable insights, our study underscores the need for robust estimation techniques and careful consideration of heterogeneity in firm-level data to discern reliable patterns in asset pricing dynamics.

4.2 Quantile regression model results

In our second hypothesis (H2), we propose that the PTV-return relationship may show heterogeneity across distinct segments of the return distribution, notably in the extreme tails (representing extremely high and low returns). Investors may depict the heterogeneous risk attitude and preferences based on the degree of perceived gain. Therefore, relying solely on estimations centered around the conditional mean (Table 3) might lead to a misleading understanding. Specifically, we propose that investors' risk evaluation and preferences may differ for securities under or over-performing the mean.

In our empirical analysis, first, we consider univariate QR models (Equation 4), and Table 4 (Panel A) reports the results. In contrast to conditional mean models, QR provides a more robust picture, and we witness an asymmetric relationship between PTV and stock

Quantiles	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95
<i>Panel A: Principal Analysis</i>											
PTV	0.1367*** (5.54)	0.1126*** (5.06)	0.0800*** (3.9)	0.0583*** (3.17)	0.0381** (2.15)	0.0203 (1.15)	−0.0027 (−0.15)	−0.0211 (−1.17)	−0.0488** (−2.45)	−0.076*** (−3.2)	−0.1161*** (−4.68)
N	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689
Goodness of fit	1.60%	1%	0.40%	0.10%	0.10%	0.10%	0.10%	0.20%	0.50%	1.50%	2.50%
<i>Panel B: Alternate PTV construction window (2 years)</i>											
two years	0.1172*** (4.88)	0.0921*** (3.84)	0.0718*** (3.57)	0.0468** (2.60)	0.0308* (1.78)	0.0157 (0.97)	−0.0002 (−0.01)	−0.0151 (−0.89)	−0.031 (−1.63)	−0.0372 (−1.54)	−0.0603** (−2.43)
N	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689
Goodness of fit	1.50%	0.80%	0.30%	0.10%	0.10%	0.10%	0.10%	0.10%	0.40%	1.20%	1.90%
<i>Panel C: Alternate PTV construction window (5 years)</i>											
five years	0.1365*** (6.34)	0.117*** (5.78)	0.0813*** (4.76)	0.0495*** (3.02)	0.0341** (2.06)	0.0144 (0.88)	−0.01 (−0.57)	−0.0271 (−1.6)	−0.065*** (−3.44)	−0.1159*** (−6.24)	−0.1681*** (−7.34)
N	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689
Goodness of fit	1.40%	1%	0.40%	0.20%	0.10%	0.10%	0.10%	0.20%	0.60%	1.50%	2.50%
<i>Panel D: Alternate Reference point (Raw Return)</i>											
RR	0.1438*** (5.78)	0.1212*** (5.10)	0.0871*** (4.00)	0.0583*** (2.88)	0.0396* (1.87)	0.0204 (1.03)	−0.0049 (−0.24)	−0.0253 (−1.3)	−0.0593*** (−2.85)	−0.102*** (−4.51)	−0.1325*** (−5.1)
N	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689
Goodness of fit	1.50%	0.90%	0.30%	0.10%	0.10%	0.10%	0.30%	0.70%	1.30%	2.70%	4.10%
<i>Panel E: Alternate Reference point (Return over Risk-free Rate)</i>											
RF	0.1439*** (5.80)	0.1203*** (5.05)	0.0866*** (3.99)	0.0578*** (2.84)	0.039* (1.84)	0.0201 (1.01)	−0.0035 (−0.17)	−0.0236 (−1.21)	−0.0581*** (−2.78)	−0.0965*** (−4.23)	−0.1326*** (−5.15)
N	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689
Goodness of fit	1.50%	0.90%	0.30%	0.10%	0.10%	0.10%	0.30%	0.70%	1.30%	2.80%	4.10%
<i>Panel F: PTV is constructed using returns from month $t-37$ to month $t-2$</i>											
Reversal	0.1433*** (6.06)	0.1161*** (5.60)	0.0858*** (4.45)	0.0626*** (3.53)	0.0404** (2.44)	0.0217 (1.36)	−0.0012 (−0.07)	−0.0206 (−1.25)	−0.0533*** (−2.76)	−0.0882*** (−3.87)	−0.1291*** (−5.12)
N	41,265	41,265	41,265	41,265	41,265	41,265	41,265	41,265	41,265	41,265	41,265
Goodness of fit	1.60%	1%	0.40%	0.20%	0.16%	0.04%	0.10%	0.20%	0.50%	1.55	2.50%
(continued)											

Table 4.

Quantiles	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95
<i>Panel G: Unfiltered Data (Variables not Winsorized)</i>											
PTV	0.1448*** (4.79)	0.1161*** (4.95)	0.0821*** (4.05)	0.0623*** (3.84)	0.0432*** (2.72)	0.0248 (1.54)	0.0038 (0.22)	−0.0165 (−0.97)	−0.0423** (−2.13)	−0.0777*** (−3.38)	−0.1211*** (−4.27)
N	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689
Goodness of fit	1.38%	0.91%	0.37%	0.14%	0.03%	0.00%	0.06%	0.18%	0.43%	1.30%	2.14%
<i>Panel H: Sample Period Since 2000</i>											
PTV	0.1189*** (5.07)	0.105*** (4.38)	0.0784*** (4.75)	0.0572*** (3.62)	0.0449*** (2.99)	0.0298** (2.12)	0.0106 (0.76)	−0.0137 (−0.93)	−0.0428*** (−2.61)	−0.0988*** (−5.53)	−0.1431*** (−7.52)
N	52,206	52,206	52,206	52,206	52,206	52,206	52,206	52,206	52,206	52,206	52,206
Goodness of fit	0.60%	0.60%	0.30%	0.10%	0.00%	0.00%	0.10%	0.20%	0.50%	1.30%	2.20%
<i>Panel I: PTV with Indian Parameters</i>											
PTV	0.086*** (3.26)	0.0599*** (2.78)	0.0496** (2.58)	0.0431** (2.45)	0.0294* (1.77)	0.0209 (1.24)	0.0157 (0.97)	0.0095 (0.58)	−0.007 (−0.38)	−0.0192 (−0.96)	−0.0409 (−1.65)
N	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689
Goodness of fit	0.78%	0.53%	0.22%	0.09%	0.04%	0.00%	0.01%	0.04%	0.13%	0.60%	1.16%
<i>Panel J: Expanded NIFTY 500 Universe</i>											
PTV	0.3479*** (12.24)	0.2796*** (11.41)	0.2137*** (9.07)	0.1772*** (8.06)	0.1404*** (6.83)	0.1091*** (5.53)	0.0673*** (3.32)	0.0261 (1.28)	−0.0282 (−1.31)	−0.1248*** (−4.47)	−0.2186*** (−6.96)
N	82,968	82,968	82,968	82,968	82,968	82,968	82,968	82,968	82,968	82,968	82,968
Goodness of fit	7.90%	5.40%	2.90%	1.60%	0.80%	0.30%	0.10%	0.01%	0.40%	2.20%	4.7%

Note(s): (1) “N” indicates the number of observations. (2) *, **, and *** indicate statistical significance at 10%, 5%, and 1% levels

The table presents the results from the QR framework [following Equation (4)], using PTV across the conditional quantiles of the return distribution. The coefficients of the PTV measure, corresponding *t*-statistics (below), and the Goodness-of-fit measure for the regression are reported. QR is executed using the “*Quantreg R package*” (Koenker *et al.*, 2018). The “*WRTDStidal package*” in R is used to compute the goodness-of-fit (GoF) measure. The computation involves subtracting one from the ratio between the appropriately weighted sum of absolute deviations from the conditional and unconditional models (Koenker and Machado, 1999)

Source(s): Created by authors

returns. Our findings reveal that PTV parameter estimates are positive for lower quantiles but tend to exhibit a negative association in upper quantiles. Coefficients for central quantiles are statistically insignificant, indicating that, on average, PTV does not significantly impact returns (like pooled OLS and Panel fixed effect estimations).

Nonetheless, as we proceed further from the median quantile toward the extremes of the return distribution, the relationship undergoes a change and becomes statistically significant. The forecasting ability of PTV increases gradually as we move from the median quantile and becomes more robust, as indicated by the increasing pseudo-adjusted R^2 values (indicating the goodness of fit) and the t-statistics associated with PTV. However, the insignificant relationship between PTV and future returns at the median quantile ($\tau = 0.5$) should be seen with caution because this insignificance may result from averaging significant contrasting findings on both tails (Bassett and Chen, 2001) [13]. With this, our study advances the extant literature that limits itself to the average relationship.

Our evidence reveals a sign reversal in the PTV-return relationship across the return distribution, supporting our hypothesis (H2) about heterogeneity in this relationship. In the upper quantiles, the PTV-return share a negative relationship. For instance, at the 95th quantile, an increase of one SD in PTV results in a 1.05% decrease in stock return [14]. Conversely, the opposite relationship is observed in the lower quantiles. At the 5th quantile, where we observe a positive relationship, one SD increase in PTV is associated with a 1.35% increase in returns. This QR approach reveals a more complex relationship between PTV and returns than previously documented by studies relying on conditional mean models, highlighting the importance of considering the entire return distribution and contributing new evidence to the literature on PT and asset pricing. Thus, it is tempting to ask the underlying behavioral explanation for this sign-reversal phenomenon across different conditional quantile distributions.

To answer this question, we argue that the segmented behavioral pattern on either side of the median quantile might indicate the investors' heterogeneous risk attitude and preferences based on the degree of perceived gains (adjusting risk preferences based on perceived gains). The inverse relationship evident in the upper quantiles is well evidenced in the extant literature using mean models. These studies propose the overvaluation hypothesis and suggest that stocks with high PTV are perceived as attractive by investors. This attractiveness leads to the overvaluation of these stocks and, consequently, lower subsequent returns. However, our analysis uncovers a distinct pattern: a positive relationship in the lower quantiles. A comparable relation (a positive average correlation between PTV and future returns) has been previously noted by Son and Nguyen (2019) and do Nascimento *et al.* (2021) in South Korean and South African markets, respectively. They designate these differences in the findings to differences in the cultures and value systems of the individuals.

Further, we note that the divergence is not only in the direction of the relationship but also in the magnitude. That is, the association between PTV and future returns not only reverses signs across different levels of returns but also exhibits varying levels of variability or spread in expected returns. We further note that the distribution of future returns is more divergent in lower return quantiles than at high returns. This means that, at lower quantiles, higher PTV values have a stronger positive impact on returns than higher PTV values, which have a negative effect on the expected returns in the upper quantile and vice versa. Thus, the PTV-return relationship is positive at lower quantiles and negative at higher quantiles, and the degree of change in returns is not consistent across all quantiles, leading to a nonlinear pattern of divergence. Additionally, more coefficients are statistically significant below the median quantile, advocating that the PTV-returns relationship is more robust and consistent in the lower half of the distribution of returns. It also indicates that PTV has a more substantial impact on returns during periods of market stress or tail events when returns are lower.

To explain the positive PTV-return relationship, our argument is consistent with investors' irrational belief that low-return stocks (below-average expected returns) possess greater upside potential (Birru and Wang, 2016). This indicates that investors tend to favor stocks with lower three-year returns (within the PTV estimation window) when they are less cautious and do not always behave as the PT advocates (PT says that investors prefer stocks with higher past mean returns). In conclusion, our findings underline the effect of behavioral finance in general and cognitive biases in particular on asset pricing (Barberis *et al.*, 2003, 2006; Barberis and Huang, 2008; Barberis and Thaler, 2003; Shiller, 2003) and add a significant predictor to the Indian equity market returns.

4.2.1 Robustness check. The definition of past return distribution is critical in exploring the predictability of PTV on security returns. We establish the validity of our results using varying time windows for the construction of PTV (2, 3, and 5 years) and examine how sensitive the PTV is to the estimation period. Table 4 presents the findings utilizing these alternative PTV metrics, along with bootstrapped standard errors and variance-covariance matrices. To ensure the reliability of our results, we also re-evaluate our model using raw return and return in excess of the risk-free rate. The results presented in Table 4 exhibit a similar relationship pattern, except at the median quantile (significantly negative), validating the consistency of our conclusions.

For further robustness, we analyze unfiltered data without winsorizing returns or variables. Findings remain unchanged, indicating winsorization has no significant impact. An analysis spanning a larger period (since 2000) yields consistent results with the principal analysis, detailed in Appendix D. To bolster our findings, we also incorporate parameters obtained from Indian participants, as reported by Rieger *et al.* (2017), into our analysis and observe that findings do not change. Moreover, our main analysis focused on current NIFTY 500 Index stocks (as of 13 May 2024). To account for historical index changes and mitigate the survivorship bias, Table 4 (Panel J) presents results for all stocks ever included during the sample period in the NIFTY 500. These results align with our main findings, confirming the study's robustness across different sample compositions.

4.3 Controlling for stock-specific variables

We expanded the basic model from Equation (4) to include additional macroeconomic and firm-specific explanatory variables and to examine whether the predictability of PTV is because of the omission of other significant variables discussed in the literature. We estimate the model detailed in Equation (5). Table 5 shows the coefficients of PTV using QR. Notably, the $\beta(\tau)$ estimates from this extended model align with the corresponding slopes observed in the univariate QR models reported in Table 4 Panel A. Therefore, our results continue to be robust after accounting for systematic and other risk factors.

We conducted additional analyses to address the concern that the short-term reversal effect might influence our findings. We demonstrated that PTV maintains its predictive ability even after accounting for short-term reversals in our regression models. Furthermore, we introduced a one-month gap between calculating PTV and measuring returns (Table 4 Panel F), and the results remained consistent. Further, we considered MAX and skewness measures in our models to ensure that the predictive strength of PTV extends beyond the currently established proxies for lottery-like characteristics.

Next, we also apply the Wald slope equality test to examine if the estimated parameters on the selected quantile (τ) differ from the median and corresponding $(1 - \tau)$ quantile and also as a robustness check for heteroscedasticity (Koenker and Bassett, 1982). The baseline hypothesis is:

$$H_0 : \beta_1(\tau_1) = \beta_2(\tau_2) = \dots = \beta_n(\tau_n)$$

Quantiles	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95
PTV	0.1142*** (6.03)	0.0972*** (6.33)	0.0753*** (5.03)	0.0512*** (3.82)	0.0385*** (2.73)	0.0236* (1.85)	0.0065 (0.57)	−0.0084 (−0.64)	−0.0263** (−2.09)	−0.0553*** (−3.31)	−0.0755*** (−3.99)
Beta	−0.0187 (−1.06)	−0.0063 (−0.47)	−0.0081 (−0.77)	−0.0119 (−1.19)	−0.0111 (−1.15)	−0.0136 (−1.3)	−0.011 (−1.01)	−0.0135 (−1.23)	−0.0215** (−2.06)	−0.0222 (−1.57)	0.006 (0.40)
Size	0.0184 (0.87)	0.0057 (0.34)	8.00E−04 (0.05)	−0.0065 (−0.47)	−0.016 (−1.15)	−0.0337** (−2.39)	−0.0578*** (−4.1)	−0.0847*** (−5.07)	−0.1139*** (−6.18)	−0.1643*** (−7.62)	−0.2096*** (−9.09)
Bm	−0.0015 (−0.1)	0.0082 (0.56)	0.0056 (0.37)	0.0031 (0.23)	0.0063 (0.44)	0.0028 (0.19)	0.0006 (0.04)	0.0014 (0.09)	0.0048 (0.27)	0.0142 (0.66)	0.0242 (1.02)
Mom	−0.0235 (−1.22)	−0.0187 (−1.29)	−0.006 (−0.54)	0.0092 (0.81)	0.0272** (2.41)	0.0337*** (2.80)	0.0459*** (3.35)	0.0693*** (4.75)	0.0791*** (4.69)	0.11*** (4.82)	0.1383*** (5.21)
Reversal	0.0044 (0.39)	0.0115 (1.14)	−0.0015 (−0.17)	0.0063 (0.77)	2.00E−04 (0.02)	0.002 (0.23)	0.0047 (0.55)	0.0097 (0.94)	0.0095 (0.92)	0.0203* (1.86)	0.0069 (0.45)
Illiq	−0.0336** (−2.46)	−0.0271*** (−2.75)	−0.0255*** (−3.07)	−0.0076 (−1.03)	0.0023 (0.28)	0.0132* (1.70)	0.0122* (1.67)	0.0122 (1.43)	0.016 (1.47)	0.0151 (1.14)	0.0122 (0.62)
Ivol	−0.2468*** (−7.01)	−0.203*** (−8.58)	−0.1352*** (−5.53)	−0.0999*** (−5.71)	−0.0791*** (−3.85)	−0.0444** (−2.11)	−0.0122 (−0.66)	0.0328* (1.79)	0.0766*** (5.14)	0.1153*** (4.56)	0.1169*** (3.89)
Max	0.0792** (2.23)	0.0441* (1.68)	0.0213 (1.01)	0.0145 (0.85)	0.0083 (0.46)	0.0002 (0.01)	−0.0089 (−0.47)	−0.0309 (−1.58)	−0.063*** (−3.6)	−0.0615** (−2.12)	−0.0729* (−1.95)
Min	−0.0828*** (−3.05)	−0.0762*** (−3.93)	−0.0657*** (−3.9)	−0.0481*** (−4.14)	−0.043*** (−4.23)	−0.0254** (−2.17)	−0.0123 (−0.84)	0.0031 (0.20)	−0.0058 (−0.28)	0.0086 (−0.34)	−0.0058 (−0.18)
Skewness	0.0172 (0.60)	0.0353** (2.09)	0.032** (2.28)	0.0251** (2.14)	0.022* (1.79)	0.0171 (1.50)	0.0181 (1.38)	0.0065 (0.46)	0.0112 (0.55)	0.0158 (0.57)	0.0206 (0.64)
Idio-skew	0.0045 (0.32)	0.0048 (0.37)	0.0071 (0.68)	0.0077 (0.99)	0.0057 (0.75)	0.0025 (0.39)	−0.0082 (−1.14)	0.00001 (0.00)	0.0098 (0.86)	0.0105 (0.80)	0.0249* (1.68)
Co-skew	−0.0203 (−1.25)	−0.024 (−1.55)	−0.0242* (−1.96)	−0.0164 (−1.44)	−0.0134 (−1.21)	−0.0076 (−0.7)	−0.0012 (−0.09)	0.0064 (0.44)	0.02 (1.28)	0.0494*** (2.94)	0.0309 (1.34)
Goodness of fit	4.40%	2.40%	1%	0.50%	0.30%	0.30%	0.60%	1.20%	2.40%	5.50%	8.70%
N	41,689	41,689	41,689	41,689	41,689	41,689	41,689	41,689.00	41,689	41,689	41,689

Note(s): Asterisks *, **, and *** show the significance of the relevant test statistic at 10%, 5%, and 1%, respectively

The table presents the results from the quantile regression framework [following Equation (5)], using PTV and other control variables across the conditional quantiles of the return distribution. We report the coefficients of PTV and other measures, corresponding *t*-statistics (below), and the Goodness-of-fit measure for the regression. QR is executed using the “*Quantreg R package*” (Koenker et al., 2018). The “*WRTDSidal package*” in R is used to compute the goodness-of-fit (GoF) measure. The computation involves subtracting one from the ratio between the appropriately weighted sum of absolute deviations from the conditional and unconditional models (Koenker and Machado, 1999)

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Looking at [Table 2D](#), we see that the coefficients differ across quantiles, thus rejecting the null of equality. This affirms our hypothesis that PTV and returns share an asymmetric relationship, with positive dependence in lower and negative dependence in upper quantiles.

Our study primarily focuses on testing PTV's forecast over a monthly horizon. However, we also explore its forecasting ability over longer timeframes. The rationale behind this investigation is that mispricing resulting from bounded rationality might only be partially corrected by arbitrageurs in the short term due to limitations in arbitrage activities and might not exist in the long term ([Ni et al., 2015](#)). Our findings, as depicted in [Table 6](#), reveal that PTV exhibits significant predictability with long-run excess market returns, stretching up to 12 months. Interestingly, the relationship becomes significant (positive) at the median for the $t+6$ horizon and beyond.

Moreover, to examine whether investment horizons impact investors' decision-making, the analyses in this paper are performed on NSE data of different frequencies: daily, weekly, and monthly [[15](#)]. Further, to establish the robustness of our findings and to see if the dependence relationship differs in BSE, another significant market in India. We conduct a similar analysis on the BSE S&P 500; results do not change and are provided in [Table D5](#) [[16](#)].

4.4 PT components

In our principal analysis, we focused on the PTV variable in its entirety. In this section, we individually explore the significance of each PTV component, specifically loss aversion (LA), probability weighting (PW), and the curvature of the value function (CC) within the context of the Indian equity market. We aim to discern whether any particular component holds a more dominant position compared to the others. To test this, we run additional models with only one component active at a time while re-estimating our QR model for different τ values. Like [Barberis et al. \(2016\)](#), we compute these PTV elements using distinct sets of parameters, with the outcomes detailed in [Table 7](#).

Notably, none of the PTV components show statistical significance at the median quantile. However, the loss aversion (LA) component, individually or in combination, offers the most statistical significance compared to other components and combinations of components across the quantiles. The results in [Table 7](#) emphasize that the primary driver of PTV's predictive power is loss aversion, as the most significant coefficients in [Table 7](#) correspond to PTV components related to loss aversion. Nonetheless, considering that factors such as the shape of the value function and probability weighting also influence the predictive pattern, it suggests that loss aversion alone is unlikely to be the sole factor driving the predictive association within PTV. The findings differ from the US equity market, where the probability weighting component was dominant ([Barberis et al., 2016](#)) and could be attributed to different sociocultural structures ([do Nascimento et al., 2021](#); [Kumar, 2009](#); [Son and Nguyen, 2019](#)).

5. Implications, limitations, and future research

Our study addresses a topical debate in investors' decision-making process by explicitly incorporating known psychological mechanisms into the modeling of these processes. It contributes to the literature by demonstrating PT's applicability in describing real-world decision-making behavior with significant financial implications. For instance, our FMB estimates indicate that an increase of one SD in PTV leads to a 0.021% increase in future return. In essence, the positive PTV-return association in the Indian market is about 1.5 times stronger than the negative relationship observed in the US stock market.

While the least square approach gives the slope coefficients at the mean, our study delves into the heterogeneity issue from the response front as it seeks to broaden the scope of

Quantiles	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95
$t+1$	0.1142*** (6.03)	0.0972*** (6.33)	0.0758*** (5.03)	0.0512*** (3.82)	0.0385*** (2.73)	0.0236* (1.85)	0.0065 (0.57)	−0.0084 (−0.64)	−0.0268** (−2.09)	−0.0553*** (−3.31)	−0.0755*** (−3.99)
GoF	4.40%	2.40%	1%	0.50%	0.30%	0.30%	0.60%	1.20%	2.40%	5.50%	8.70%
$t+2$	0.1038*** (5.73)	0.0979*** (6.21)	0.0802*** (5.89)	0.0627*** (4.89)	0.0484*** (4.10)	0.0326*** (3.02)	0.0113 (0.98)	−0.0066 (−0.55)	−0.0272* (−1.89)	−0.0462*** (−2.77)	−0.0734*** (−3.71)
GoF	3.40%	2.10%	1%	0.50%	0.30%	0.50%	0.90%	1.50%	2.70%	5.80%	8.70%
$t+3$	0.1087*** (6.45)	0.1038*** (7.42)	0.0732*** (5.10)	0.0527*** (5.08)	0.0383*** (3.63)	0.0236** (2.39)	0.0084 (0.75)	0.0021 (0.20)	−0.0145 (−1)	−0.0408** (−2.09)	−0.0643*** (−2.64)
GoF	3.20%	2.10%	1.10%	0.50%	0.30%	0.40%	0.80%	1.30%	2.60%	5.40%	8.30%
$t+6$	0.1061*** (6.61)	0.1024*** (6.98)	0.0874*** (8.10)	0.0667*** (7.69)	0.0518*** (8.83)	0.0400*** (3.77)	0.0228** (2.10)	0.0061 (0.52)	−0.0119 (−0.86)	−0.0438** (−2.61)	−0.0767*** (−3.65)
GoF	3.70%	2.40%	1.40%	0.90%	0.70%	0.70%	1%	1.60%	2.60%	5.20%	8.10%
$t+9$	0.1149*** (5.52)	0.0998*** (5.40)	0.0834*** (6.44)	0.0617*** (5.60)	0.0517*** (5.23)	0.0307*** (3.70)	0.0288** (2.54)	0.0123 (0.87)	0.0002 (0.01)	−0.0399*** (−2.7)	−0.0715*** (−4.75)
GoF	2.80%	2.30%	1.50%	1.10%	1%	1%	1.30%	1.80%	3%	5.80%	8.70%
$t+12$	0.1065*** (5.89)	0.0879*** (5.67)	0.077*** (5.69)	0.0643*** (4.97)	0.0518*** (3.73)	0.0373*** (2.73)	0.0203 (1.55)	0.0025 (0.18)	−0.0079 (−0.52)	−0.0396** (−2.28)	−0.0596*** (−2.74)
GoF	3.30%	2.60%	1.50%	1%	0.80%	0.80%	1.10%	1.50%	2.50%	5%	7.80%

Note(s): Asterisks *, **, and *** show the significance of the relevant test statistic at 10%, 5%, and 1%, respectively

The table presents the results from the quantile regression framework, using PTV across the conditional quantiles of the return distribution at different forecast horizons ($t+1$, $t+2$, $t+3$, $t+6$, and $t+12$). We report the coefficients of the PTV measure, corresponding t -statistics (below), and the Goodness-of-fit estimate for the regression. QR is executed using the “*Quantreg R package*” (Koenker *et al.*, 2018). The “*WRTDSidal package*” in *R* is used to compute the goodness-of-fit (GoF) measure. The computation involves subtracting one from the ratio between the appropriately weighted sum of absolute deviations from the conditional and unconditional models (Koenker and Machado, 1999)

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Table 7.
PTV components

Quantiles	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95
PTV	0.1367*** (5.54)	0.1126*** (5.06)	0.08*** (3.90)	0.0583*** (3.17)	0.0381** (2.15)	0.0203 (1.15)	−0.0027 (−0.15)	−0.0211 (−1.17)	−0.0488** (−2.45)	−0.076*** (−3.2)	−0.1161*** (−4.68)
LA	0.1326*** (5.16)	0.1034*** (4.39)	0.0797*** (3.68)	0.0597*** (3.01)	0.0431** (2.20)	0.0252 (1.38)	0.006 (0.33)	−0.0143 (−0.75)	−0.0397** (−2.17)	−0.0701*** (−3.02)	−0.1073*** (−4.63)
PW	0.0063 (0.25)	−0.0008 (−0.04)	−0.0002 (−0.01)	0.0018 (0.11)	0.0066 (0.39)	0.0114 (0.63)	0.0234 (1.27)	0.0363* (1.90)	0.0496** (2.57)	0.0772*** (3.26)	0.0784*** (2.99)
CC	0.0308 (1.15)	0.027 (1.18)	0.0232 (1.13)	0.0238 (1.28)	0.0246 (1.33)	0.0208 (1.13)	0.025 (1.44)	0.0325* (1.82)	0.0364* (1.92)	0.0341 (1.56)	0.0286 (1.16)
LACC	0.1264*** (4.77)	0.0999*** (4.27)	0.0769*** (3.56)	0.058*** (2.96)	0.0408** (2.10)	0.0261 (1.40)	0.0062 (0.35)	−0.0104 (−0.56)	−0.0339* (−1.89)	−0.0635*** (−3.03)	−0.0978*** (−4.34)
LAPW	0.129*** (5.26)	0.1099*** (4.95)	0.0745*** (3.74)	0.0563*** (3.18)	0.035** (2.02)	0.0179 (0.99)	−0.002 (−0.11)	−0.02 (−1.12)	−0.0448** (−2.24)	−0.0705*** (−3.08)	−0.1031*** (−4.35)
CCPW	0.0181 (0.67)	0.0068 (0.33)	0.0059 (0.32)	0.008 (0.47)	0.0086 (0.49)	0.0141 (0.75)	0.0218 (1.19)	0.036* (1.87)	0.0474** (2.43)	0.0666*** (2.83)	0.0726*** (2.79)

Note(s): Asterisks *, **, and *** show the significance of the relevant test statistic at 10%, 5%, and 1%, respectively. The table presents the results from the quantile regression framework using PTV, PTV components, and combinations of components across the conditional quantiles of the return distribution. We report the coefficients of the PTV measure, corresponding *t*-statistics (below), and the Goodness-of-fit estimate for the regression. QR is executed using the “*Quantreg R package*” (Koenker *et al.*, 2018). The “*WRTDStidal package*” in *R* is used to compute the goodness-of-fit (GoF) measure. The computation involves subtracting one from the ratio between the appropriately weighted sum of absolute deviations from the conditional and unconditional models (Koenker and Machado, 1999).

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findings by employing the QR method. It offers a more complete representation of the PTV-return relationship than estimates from the typically used mean models. This approach offers constructive insights and suggestions to investors to improve the efficacy of their investment strategies. While the PTV-return relationship is insignificant at the median, the findings depict high economic and statistical significance at lower and higher quantiles. For instance, at the 95th quantile, an increase of one SD in PTV results in a 1.05% decrease in stock return, significantly larger than the mean-level effect observed in the US market. For portfolio managers, this distributional information from QR broadens the range of investment opportunities. In this regard, [Ma and Pohlman \(2008\)](#) establish that forecast and portfolio construction based on distribution information from QR yields more precise forecasts and potentially higher value-added portfolios. Our analysis also suggests that future research employing QR within linear asset pricing models is promising, especially in extreme market conditions for portfolio construction. In this context, future studies can explore how the distributional information from QR can be utilized to provide more realistic forecasts and create a more optimal portfolio.

Policymakers can benefit from understanding these dependence patterns (between return and PTV) to anticipate market movements and manage volatility arising from investor behavior during extreme market conditions. Further, policymakers can also utilize these findings to educate investors on heuristics and biases influencing their investment decisions and their implications for their financial well-being.

However, it is important to recognize that our analysis primarily banks on empirical evidence and, therefore, lacks any direct data on the genuine parameter levels assessed from the market and investors. Consequently, we use an indirect test to examine a cross-section of stock returns instead of observing investor actions directly. Moreover, this approach does not account for the fact that different investors may have heterogeneous behavior ([Shapira and Venezia, 2001](#)). Thus, forthcoming research should endeavor to offer more direct experimental evidence about the PTV-return relationship, offering a promising path for further investigation.

In addition, we expect future research to integrate PTV with other observable factors related to behavioral finance, like market uncertainty, investor attention, and sentiment, and examine whether these factors have any moderating effect on the PTV-return relationship. Another interesting extension of our study would involve incorporating elements from other behavioral theories, such as salience theory ([Bordalo et al., 2012](#)) and regret theory ([Bell, 1982](#)), to model the decision-making process more realistically.

6. Conclusion

Much of the research on the factors influencing stock returns typically operates under the assumption that investors are expected utility maximizers. However, with increasing evidence of deviations in the behavior, the PT has gained popularity as an alternative framework to explain decision-making in situations involving risk and uncertainty. Using the QR approach within the FMB procedure, we investigate whether PT can provide insights into these dynamics.

Our main empirical finding is the asymmetric behavioral pattern emerging on either side of the median quantile. This pattern reveals investors' varying risk attitudes and preferences based on their perception of potential gains. Notably, in the Indian equity market, we find a prevalence of loss aversion behavior, in contrast to the lottery preferences observed among US investors ([Barberis et al., 2016](#)), which can be attributed to the different socio-economic structures of Indian society ([Kumar, 2009](#)). The results we present hold substantial economic significance and are of great importance to both professionals and scholars in the field.

In summary, our research makes a substantial contribution to the field of stock cross-section analysis and behavioral finance literature by providing valuable insights into key behavioral dynamics and their relationship to stock returns. This deepens our understanding of stock price dynamics and their underlying behavioral drivers.

Notes

1. We also undertake our analysis on pooled QR; results remain qualitatively similar.
2. "<https://www.moneycontrol.com/news/business/earnings/with-3-31-trillion-valuation-india-reclaims-fifth-spot-in-global-stock-market-tally-10685511.html>"
3. Retail investors constituted 40% of the Indian stock market. While not a majority, this percentage represents a significant portion of the market, especially when compared to other major global markets where retail participation is often lower ("<https://www.capitalmind.in/2021/05/omg-the-retail-investor-is-the-biggest-player-in-the-stock-market/>").
In the USA, retail investors accounted for approximately 20% of equity trading volume in 2020 (<https://www.forbes.com/sites/forbesagencycouncil/2022/11/04/the-rise-of-the-retail-investor/>). In Brazil, a comparable developing economy, the total trading value has about 25% retail market share (Gurrola-Perez *et al.*, 2022).
4. The influence of PT is more robust for stocks with fewer institutional investors (Barberis *et al.*, 2016). This is because less sophisticated individual investors are more inclined to assess a stock's appeal by relying on past performance indicators like price charts and technical trading strategies.
5. Hou *et al.* (2020) argue that since finance is mostly observational in nature, unlike the natural sciences, there is a need to check the credibility of findings against 'similar, but not identical' specifications. In their study, they conclude that the majority of anomalies reported in the financial market fail to replicate.
6. Prior research primarily relied on monthly data to analyze PTV and cross-sections of stock returns. However, recent advancements in information technology have made daily and weekly data easily accessible (Engle, 2000). Utilizing these finer time intervals provides several advantages, including increased data points, which in turn yield more efficient estimations. Additionally, the prevalence of High-Frequency Trading (HFT) has led to swift corrections of market anomalies within short time horizons (O'Hara, 2015). Therefore, it is advisable to test for mispricing at the highest possible frequency to capture and analyze these rapid market adjustments effectively.
7. According to Kumar (2009), lottery preferences are affected by a range of socioeconomic and psychological factors. Therefore, the reduced significance of the probability weighting element of PTV (which reflects lottery preferences) may be attributed to the conservative societal norms in India, where participation in lotteries is not as prevalent as in the US.
8. Using QR information, Ma and Pohlman (2008) present two comprehensive approaches for predicting returns and constructing portfolios. They illustrate that these approaches yield more precise forecasts and lead to portfolios with greater added value compared to the conditional mean models.
9. We have also included Appendix B, which summarizes the important literature on the cross-section relationship of PTV and several asset classes.
10. They also establish the robustness of their findings at 48 and 72 months.
11. "Risk-free rate is the yield on 91-day Treasury bills provided by the Reserve Bank of India (RBI)".
12. We test for heterogeneity in our data with the help of various panel data tests: pooltest, pFtest, pWtest, plmtest, etc., and reject the null of no heterogeneity.
13. Realizing the limitations of defining a portfolio's style with a single measure, Bassett and Chen (2001) employ QR to capture the distributional perspective of portfolio management styles. They state that "substantial positive and negative impacts in the tails can counterbalance each other, resulting in only marginally significant outcomes from the conditional mean estimates".

14. The coefficient on unscaled PTV at the 95th quantile is -0.2566 , with an average cross-sectional SD of PTV is 0.041 . Thus, the effect size is -1.05% ($= -0.2566 * 0.041 * 100$).
15. The additional results related to PTV construction window of 60 months are provided in [Appendix C: 1C and 2C](#).
16. Although the NSE Nifty 500 and BSE S&P 500 share 90% of their constituents, the variations in weighting schemes (such as free-float market capitalization in NSE versus full market capitalization in BSE) and different trading environments, offers the compelling motivation for a robustness check. Furthermore, replicating the analysis across both indices strengthens the reliability and generalizability of the findings across India's two primary exchanges.

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Further reading

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Supplementary material

The supplementary material for this article can be found online.

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